

Cycles 1-2 Clone 1 Report — M-GEN-1/palette-driven-batch-v2-sampler-diversified (Fork 07063458736e)

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[OUTPUT: report_cycles_1-2_clone_1]

Cycles 1-2 Clone 1 Report — M-GEN-1/palette-driven-batch-v2-sampler-diversified (Fork 07063458736e)

Abstract

Cycles 1-2 of clone-1 (fork 07063458736e) close the cycle-35 Branch B sampler-side diversification response to c34 BATCH_SPREAD_COLLAPSED at **SPREAD_STILL_COLLAPSED** — a first-class negative finding under the pre-registered 3-verdict rubric. Cycle 1 executed the full pipeline and produced a mechanism-precise diagnosis: the c33 render_stem(stem, instrument, out_dir) API surface **never consumes pinned_state**, so even validated per-salt-distinct rule triples and distinct v2 pinned-state payloads collapse to a single byte-identical bare_combined.wav. Cycle 2 is c30-codified re-invocation-as-verification standby (single-paragraph declaration; zero writes; auditor decision **COMPLETE**). Handoff to c36 as M-GEN-1/palette-driven-batch-v3 with Options A (extend render_stem) and B (author c33-peer VST3 renderer).

Verdict

SPREAD_STILL_COLLAPSED (rubric-satisfied first-class negative finding; VALIDATED at cycle 1; **COMPLETE** at cycle 2).

Rubric SHA Anchor Chain

Location	SHA-256
docs/palette_driven_batch_v2_sampler_diversified_rubric.md	749973025e5fa4c18c745eb0aedfda3773be003b72eab390754ba21e18aae
data/gen_palette_batch_v2/rubric_hash.txt	749973...aeb6c
verdict.json.rubric_hash	749973...aeb6c

Byte-equal across all three locations; rubric-before-scripts mtime ordering enforced by test.

Mechanism-Precise Diagnosis (Load-Bearing Finding)

The c33 render dispatcher signature `render_stem(stem, instrument, out_dir)` does not accept or consume the `pinned_state` field. Even when the assignment builder correctly diversifies per-salt rule triples and computes distinct v2 iterated_params pinned-state payloads (via deterministic per-rule perturbation from SHA-256 of $(rule_id, param_name) \rightarrow$ typed delta table; no PRNG), the pinned-state information is discarded at the render-dispatch boundary. Metadata-layer diversification is real; audio-layer diversification is impossible without changing the render API.

Metadata-layer diversification empirically confirmed at cycle 1: - **3 distinct assignment SHAs** across salts 0/1/2. - **3 distinct rule-id triples** across salts (via SHA-256 tiebreak over full K=20 harmonic + K=18 rhythmic + K=15 arrangement pre-i3 pools). - **6 distinct v2 payload SHAs** (3 salts $\times$ 2 instruments Surge XT + Dexed; drums remain fluidsynth-static per spec).

Audio-layer collapse: universal `bare_combined.wav` SHA `a8c1557c...2ba794` cross-salt (identical to the c33 single-seed anchor), with `IQR = 0` and `max-min = 0` on 4 numeric-family keys $\times$ 2 panels (both `panel_original` and `panel_fluidsynth`).

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	Ship the milestone under frozen 3-verdict rubric	Full pipeline; SPREAD_STILL_COLLAPSED with mechanism structurally characterized; ledger events emitted	VALIDATED
2	Verification-only standby	Single-paragraph declaration; zero writes; canonical rubric SHA cited; no re-computation, no re-emission	COMPLETE (no-null-cycle rule satisfied)

Cycle 2 respected all 12/12 verification-only criteria: no new files; no re-computation; no re-rendered/re-measured artefacts; no re-emitted ledger events; no writes under `docs/`,

scripts/gen_palette_batch_v2/, data/gen_palette_batch_v2/, or tests/; no perturbation-surface tweak attempts; no extension of c33 render_stem or c33-peer VST3 renderer author (both c36 scope); no read of ledger_i3_dminor.jsonl or import of i4_stratified.py; no read of sibling clone-0/clone-2 shadow ledgers; no new mechanism claim or falsification criterion.

Test Surface (Established at Cycle 1)

Suite	Result
tests/test_palette_driven_batch_v2.py	20/20 PASS (exceeds ≥ 14 minimum)
tests/test_integration_cross_branch.py §55	PASS ; whole suite 0 failures
python3 -m long_exposure.tools.promise_check .	0 ERRORS , 114 WARN (merge-clearable or established-exemption)

Per-salt byte-determinism $\times 2$ holds across salts 0/1/2 (each salt's two runs into fresh temp-file.mkdtemp() dirs yield byte-identical WAV). Per-salt distinctness FAILS at audio bytes (the rubric's SPREAD_STILL_COLLAPSED trigger); metadata layer distinctness PASSES.

Anchor Preservation

anchor_preservation.json: unchanged=True across all five READ-ONLY anchor families — c34 palette_v2; c33 palette_render; c33 dawdreamer_state; c31 palette_v1; c26/c27/c28/c29/c30 analytical utilities; c22 stability harness. c9 chain, c13 pipeline, c15 i4_stratified.py not imported (grep-verified).

Ledger Events (Cycle 1: 8 shadow rows under -clone-1 suffix; Cycle 2: 0)

Six named + two housekeeping, per spec:

1. _run/cycle_35_launched-clone-1 (status: validated per c35 Branch C newly-codified convention)
2. _plan/palette_driven_batch_v2_sampler_diversified_rubric_frozen-clone-1
3. _infra/egress-probe-cycle-35-clone-1
4. M-GEN-1/palette-driven-batch-v2-sampler-diversified (in-progress; M-* unsuffixed per c32)
5. M-GEN-1/palette-driven-batch-v2-sampler-diversified (validated verdict roll-up, SPREAD_STILL_COLLAPSED)
6. _run/cycle_35_closed-clone-1
7. _archive/cycle-35-scratch-clone-1
8. _infra/adopt-cycle35-tests-clone-1

Cycle 2: zero. validated $\rightarrow$ in_progress forbidden per c29 lesson. No M-EAR-1/* events (armed harness stays dormant per spec).

State-Machine Discipline (c29 Lemma Respected)

M-GEN-1/palette-driven-batch-v2-sampler-diversified is a peer sub-milestone under M-GEN-1. NOT a child of terminal-validated M-GEN-1/palette-driven-batch-v1 or M-GEN-1/batch-v{1..6}.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908.

- SHA-256 tiebreak; no PRNG (AST-verified); no sidcar_nonfactor imports.
- Interpreter guard assert `sys.executable == '/usr/bin/python3'` on every new script.
- Read-only anchors preserved (see Anchor Preservation above).
- Rated audio egress-blocked at `*.googlevideo.com` (unchanged 403 from c34 baseline). M-EAR-1 armed-not-fired posture holds.
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence:{level,rationale,asses
UUID5 content-hash event_id.

Anti-Patterns Locked (5-Count Stable)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted.

Cycle-36 Handoff (Published in Report; Reproduced Here)

New peer sub-milestone: M-GEN-1/palette-driven-batch-v3 under M-GEN-1 (c29 lemma respected; NOT a child of terminal-validated palette-driven-batch-v{1, 2-sampler-diversified} or batch-v{1..6}).

- **Option A (minimal audio-surface change):** extend c33 `scripts.palette_render.render_stem` to consume `pinned_state.parameter_dict` for `fluidsynth/sfizz` dispatchers; thread per-rule parameters through the existing CLI invocation. Uses this cycle's validated v2 payloads at `data/gen_palette_batch_v2/per_song/<salt>/v2_perturbed/{surge_xt,dexed}.json` as ready-made inputs.
- **Option B (strategic, palette-v2 payoff):** author a new c33-peer renderer for Surge XT / Dexed VST3 consuming `v2_iterated_params`. Subject to c35 Branch A's `RENDER_FAILS VST3 nondeterminism finding` — may push behind auditor's alternate M-DAW-SPIKE-1/{dexed-preset-hydration, surge-xt-fxp-load, vst3-render-nondeterminism-characterization} line.
- **Preserve locks:** $\alpha = 0.7469387071101908$; NO PRNG; SHA-256 tiebreak only; 5 anti-patterns locked; `get_state()/save_state(filepath)/set_state(bytes)` remain c31 STILL_GAP surface; `ledger_i3_dminor.jsonl` NOT read from this branch's descendants.
- **Merge-time housekeeping** (root-conductor scope): 114 orphan-artifact WARNs clear via `_infra/adopt-fanout-artifacts-cycle35-batch-v2-clone-1`; sibling artefact clusters reconcile under their own adoption events.

Merge Disposition

Merge report at `/home/user/music-gen-instance/fork-07063458736e/clone-1/merge_report.md` (spec path; workspace-root fallback also acceptable per c31/c34 pattern). Eight shadow-ledger rows (cycle 1) ready for `concat_clone_ledgers`; zero cross-clone collisions under c32 -clone-1 suffixes. Cycle 2 contributes zero shadow rows.

Cumulative Progress

M-GEN-1 palette line — three-cycle mechanism-focused convergence chain (textbook honest-negative-finding chain):

Cycle	Milestone	Verdict	Structural Finding
c33	M-TEX-1/palette-driven-bare-render	PALETTE_MOVES_PANEL	Palette contract activates on real renders (single-song).

Cycle	Milestone	Verdict	Structural Finding
c34	M-GEN-1/palette-driven-batch-v1	BATCH_SPREAD_COLLAPSED	Dispatcher build_assignment_row is rule_id-invariant.
c35 (this)	M-GEN-1/palette-driven-batch-v2-sampler-diversified	SPREAD_STILL_COLLAPSED	Fixing the dispatcher alone doesn't move audio bytes: c33 render_stem API surface never consumes pinned_state.

Each cycle carries a specific, falsifiable, pre-registered rubric, converging on the exact API surface that must change in c36.

Pattern durability: seven consecutive cycles of rubric-first pre-registration discipline (c26-c30 mechanism probes + c31/c32/c33/c34/c35). Zero rubric-edit-after-analysis incidents. SPREAD_STILL_COLLAPSED is the first-class negative finding pre-registration is designed to accept — not a rubric failure.

c29 state-machine lemma respected: every c35 fanout branch (A: palette-schema-v2-hydration-render; B: this — palette-driven-batch-v2-sampler-diversified; C: _infra/anchor-manifest-v1) is a NEW peer sub-milestone, not a child of terminal-validated ancestors. Ledger topology stays a DAG, not a lineage tree.

c32 fanout-namespace convention held under c33 harness-clone-namespace-guard: infra families auto-suffixed -clone-1, substantive M-* unsuffixed — no LedgerConcatError risk at merge.

M-EAR-1 armed-harness Path B: dormant/armed pending audio-egress unblock (still 403 as of c35; retry per policy is non-blocking). **Collision-modeling arc:** closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

[END OUTPUT]